

MScFE 610 FINANCIAL ECONOMETRICS

Group Work Project # 1

Problem 1

Assume that the real regression model for y is the following:

$$y(i) = \alpha + \beta x(i) + \gamma w(i) + \delta z(i) + \epsilon(i) \quad (1)$$

However, the analyst erroneously estimates the model:

$$y(i) = \alpha + \beta x(i) + \gamma w(i) + \mu(i) \quad (2)$$

Question 1a

If the error terms $\epsilon(i)$ satisfy the standard assumptions, does it mean that the error terms $\mu(i)$ do as well? Explain why or why not using rigorous formulae.

Solution :-

No, the standard assumptions for $\epsilon(i)$ do not necessarily hold for $\mu(i)$.

The true model is $y(i) = \alpha + \beta x(i) + \gamma w(i) + \delta z(i) + \epsilon(i)$

but $z(i)$ is omitted in the estimated model. Therefore, its effect becomes part of the new error term:
 $\mu(i) = \delta z(i) + \epsilon(i)$.

Now consider the conditional expectation of $\mu(i)$:

$$E[\mu(i) \mid x(i), w(i)] = \delta E[z(i) \mid x(i), w(i)] + E[\epsilon(i) \mid x(i), w(i)].$$

If the original error satisfies the standard assumption

$$E[\epsilon(i) \mid x(i), w(i), z(i)] = 0$$

then we get

$$E[\mu(i) \mid x(i), w(i)] = \delta E[z(i) \mid x(i), w(i)]$$

This does not necessarily equal zero. If $z(i)$ is related to $x(i)$ or $w(i)$, then $\mu(i)$ can be correlated with the explanatory variables.

Therefore, even if $\epsilon(i)$ satisfies the standard assumptions, $\mu(i)$ may not satisfy them because the omitted variable $z(i)$ has become part of the error term.

Question 1b

Assume that we estimate model (2) and that x and w are orthogonal. What are the estimated $\hat{\beta}$ and $\hat{\gamma}$?

Solution :-

We estimate the following model:

$$y(i) = \alpha + \beta x(i) + \gamma w(i) + \mu(i)$$

The question states that x and w are orthogonal. Therefore,

$$\sum_i x(i)w(i) = 0$$

Because of this condition, the OLS estimates of β and γ can be calculated separately. The estimate of β is $\hat{\beta} = \sum_i x(i)y(i) / \sum_i x(i)^2$,

while the estimate of γ is

$$\hat{\gamma} = \sum w(i)y(i) / \sum w(i)^2$$

Thus, when x and w are orthogonal, the estimated coefficients are

$$\hat{\beta} = \sum x(i)y(i) / \sum x(i)^2$$

$$\text{and } \hat{\gamma} = \sum w(i)y(i) / \sum w(i)^2$$

These are the estimated coefficients from model (2) when x and w are orthogonal.

Question 1c

Determine explicitly the bias for $\hat{\gamma}$ as estimated from model (2) and using the true model (1) for y . [Note: x and w are assumed orthogonal. Nothing is said about the relationships of x and w with z .]

Solution :-

From Question 1b, we have

$$\hat{\gamma} = \sum w(i)y(i) / \sum w(i)^2$$

Using the true model,

$$y(i) = \alpha + \beta x(i) + \gamma w(i) + \delta z(i) + \epsilon(i)$$

we substitute this into the expression for $\hat{\gamma}$:

$$\hat{\gamma} = \sum w(i)[\alpha + \beta x(i) + \gamma w(i) + \delta z(i) + \epsilon(i)] / \sum w(i)^2$$

Expanding the numerator gives

$$\hat{\gamma} = \alpha \sum w(i) + \beta \sum w(i)x(i) + \gamma \sum w(i)^2 + \delta \sum w(i)z(i) + \sum w(i)\epsilon(i) / \sum w(i)^2$$

Since x and w are orthogonal,

$$\sum w(i)x(i) = 0$$

Also, under the standard OLS assumptions, the error term does not contribute systematically to the estimate. Therefore,

$$\hat{\gamma} = \gamma + \delta \sum w(i)z(i) / \sum w(i)^2$$

Hence, the bias in $\hat{\gamma}$ is

$$\text{Bias}(\hat{\gamma}) = \hat{\gamma} - \gamma = \delta \sum w(i)z(i) / \sum w(i)^2$$

Therefore, the estimate of γ will be biased when the omitted variable $z(i)$ is related to $w(i)$ and δ not equals to 0. If $w(i)$ and $z(i)$ are uncorrelated, the bias term becomes zero.

Question 1d

In what particular case would the estimates of the parameter of the model α , β , and δ obtained from model (2) be the same as those obtained from model (1)?

Solution:-

In model (1), we include $z(i)$ as one of the explanatory variables, while in model (2), $z(i)$ is left out. Therefore, we need to check whether leaving out $z(i)$ changes the estimates of the other coefficients.

From Question 1c, the bias in the estimate of γ is

$$\text{Bias}(\hat{\gamma}) = \hat{\gamma} - \gamma = \delta \sum w(i)z(i) / \sum w(i)^2$$

For this bias to be zero, we need

$$\sum w(i)z(i) = 0$$

This means that $w(i)$ and $z(i)$ have no linear relationship. In simple terms, changes $z(i)$ in should not be associated with changes in $w(i)$.

Similarly, for the estimate of β to remain unchanged, we need

$$\sum x(i)z(i) = 0$$

This means that $x(i)$ and $z(i)$ should also be uncorrelated.

Therefore, the estimates of the coefficients in the two models will be the same when the omitted variable $z(i)$ is uncorrelated with both $x(i)$ and $w(i)$

$$\sum x(i)z(i) = 0 \text{ and } \sum w(i)z(i) = 0$$

In this situation, although $z(i)$ has been left out of model (2), its omission does not affect the estimates of the coefficients of $x(i)$ and $w(i)$.

Question 1e

Using simulations, create a known regression model containing two predictors. In this case the parameters of the model are known. Now omit one of the predictors and estimate the model. Compare the estimates of the coefficient of the non omitted predictor in both circumstances. What do you notice? If you increase the sample size of the simulation, does the result change? (Hint: You create a model like

$$Y(i) = a + bX(i) + cZ(i) + e(i)$$

where a , b , c are known. Try simulating X and Z so that they are not necessarily independent and then estimate the model with 2 and only 1 predictor and pay attention to the estimates for b without omitting Z and omitting Z . e should be independent of X and Z , however!!)

Solution :-

The question suggests creating a regression model of the form

$$Y(i) = a + bX(i) + cZ(i) + e(i)$$

where a, b, and c are known parameters. It also asks us to simulate X(i) and Z(i) so that they do not necessarily have to be independent, while e(i) should be independent of both X(i) and Z(i) .

For the simulation, I choose the following values:

$$a=2, b=1.5, c=2.$$

Therefore, the specific model used in the simulation is

$$Y(i) = 2 + 1.5X(i) + 2Z(i) + e(i)$$

Here, the true coefficient of X(i) is 1.5. We will use this value as the benchmark when comparing the estimated coefficients.

To make X(i) and Z(i) correlated, we generate Z(i) first and then generate X(i) using Z(i). The error term e(i) is generated independently.

Model 1 — Both predictors included

$$Y(i) = \alpha + \beta X(i) + \gamma Z(i) + e(i)$$

Model 2 — Z(i) omitted

$$Y(i) = \alpha + \beta X(i) + u(i)$$

Here, Z(i) is deliberately left out.

We then compare the estimated coefficient of X(i) from the two models. We also repeat the simulation with different sample sizes to see whether increasing the sample size changes the result.

The sample sizes used are 100, 500, 1,000, and 5,000. The estimated coefficients obtained from the two models are shown below.

Simulation Results

Sample size	β from Model 1 (X and Z)	β from Model 2 (only X)
100	1.3667	2.7832
500	1.4599	2.9376
1,000	1.4873	2.8853
5,000	1.5137	2.9273

The true value of the coefficient of X(i) is 1.5. From the table, when both X(i) and Z(i) are included in the model, the estimated coefficient of X(i) is close to 1.5 and becomes closer to the true value as the sample size increases.

However, when Z(i) is omitted, the estimated coefficient of X(i) is much higher, around 2.8 to 2.9. Increasing the sample size does not make this estimate approach the true value of 1.5.

This happens because Z(i) is related to X(i) and also affects Y(i) . Therefore, when Z(i) is left out of the regression, part of its effect is incorrectly captured by the coefficient of X(i).

Problem 2

Question 2a

Address the sensitivity to outliers for regression models in general. About half a page to a page in length, no need for more!

Solution :-

Outliers can have a significant effect on regression models, especially when using ordinary least squares (OLS). An outlier is an observation that is very different from the other observations in the dataset.

In OLS regression, the coefficients are estimated by minimizing the sum of the squared residuals:

$$\alpha, \beta_{\text{min}} = \arg \min_{\alpha, \beta} \sum_{i=1}^n (y(i) - \alpha - \beta x(i))^2$$

Because the residuals are squared, an observation with a very large error can have a much larger effect on the regression than a normal observation. As a result, even a small number of outliers can change the estimated slope and intercept.

The effect can be stronger when an outlier has an unusual value of the explanatory variable. Such an observation has high leverage and can pull the fitted regression line toward itself. This can change the estimated relationship between the variables and may also affect measures such as the standard errors.

However, an outlier should not simply be removed from the dataset. It is important to first check why the observation is unusual. It could be due to a data-entry or measurement error, but it could also be a genuine observation.

Therefore, checking for outliers is an important part of regression analysis. Residual plots and other diagnostic measures can be used to identify observations that have a strong influence on the regression results.

Question 2b

Then, use a simulated instance to discuss/illustrate how the presence of outliers can change the estimated parameters for a regression model. Make a special effort to describe clearly how your simulation project was put together.

Solution :-

For this part, I used a simple simulated regression model to see how an outlier can affect the estimated regression parameters. The idea is to first create a dataset without any unusual observations and estimate the regression line. Then, I add a few outliers and estimate the model again to compare the results.

I started with the following regression model:

$$Y(i) = 2 + 3X(i) + e(i)$$

Here, the true intercept is 2 and the true coefficient $X(i)$ of is 3. I generated the values of $X(i)$ from a normal distribution and generated the error term $e(i)$ independently.

First, I estimated the regression model using the original data. After that, I added a few observations that were far away from the general pattern of the data. I then estimated the same regression model again.

The purpose of this simulation is to compare the estimated intercept and slope before and after adding the outliers. If the regression is sensitive to the outliers, the estimated parameters should change after the unusual observations are added.

The simulation therefore follows these steps:

- Generate the original $X(i)$ and $Y(i)$ observations.
- Estimate the regression model and record the estimated parameters.
- Add a small number of outlier observations.
- Estimate the regression model again.
- Compare the estimated parameters before and after adding the outliers.

Simulation Results

Model	Estimated Intercept	Estimated Slope
Without outliers	2.0253	2.9885
With outliers	1.2675	-1.1041

Without the outliers, the estimated intercept is **2.0253** and the estimated slope is **2.9885**. These values are very close to the true values of 2 and 3 used in the simulation.

After adding the three outliers, the estimated intercept changes to **1.2675**, while the estimated slope changes substantially to **-1.1041**. In particular, the slope changes from being close to 3 to becoming negative.

This shows that the outliers have a strong effect on the OLS estimates. Even though only three observations were added to the original 100 observations, they were far from the main pattern of the data and therefore had a large influence on the fitted regression line.

Overall, the simulation shows that OLS regression can be sensitive to outliers. A small number of unusual observations can change the estimated regression parameters considerably, especially when the observations have extreme $X(i)$ values.

Question 2c

Suppose that you have a dataset consisting of n observations $(x_1, y_1), \dots, (x_n, y_n)$. You are fitting a simple linear regression model:

$$Y_t = \alpha + \beta x_t + \epsilon_t$$

Imagine that $n-1$ points are clustered tightly along the line $y=0$, that is $(x_i, y_i) = (x_i, 0)$ for $i=1, 2, \dots, n-1$. Now, introduce a single outlier at (x_n, y_n) where x_n is very large (a high leverage point).

Prove that as $x_n \rightarrow \infty$, the OLS estimate for the slope is driven entirely by the ratio y_n/x_n regardless of the other $n-1$ points.

Solution :-

We have the simple regression model $Y_t = \alpha + \beta x_t + \epsilon_t$

According to the question, the first $n-1$ observations lie on the line $y=0$. Therefore,
 $y_i = 0, i=1,2,\dots,n-1$

The last observation is the outlier (x_n, y_n) , where x_n is very large.

The OLS estimate of the slope is:

$$\hat{\beta} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Since the first $n-1$ values of $y(i)$ are zero, the only observation that has a non-zero y -value is the outlier. Therefore,

$$\bar{y} = \frac{y_n}{n}$$

As x_n becomes very large, the outlier has a much larger effect on the calculation of the slope than the other observations. The numerator of the OLS estimate is therefore mainly determined by the outlier and is approximately proportional to $x_n y_n$

Similarly, the denominator is mainly determined by the large value of x_n and is approximately proportional to x_n^2

Therefore, when x_n is very large, the OLS slope can be approximated by

$$\hat{\beta} \approx \frac{x_n y_n}{x_n^2}$$

which simplifies to $\hat{\beta} \approx \frac{y_n}{x_n}$

Therefore, as x_n approaches infinity,

$$\lim_{x_n \rightarrow \infty} \hat{\beta} = \frac{y_n}{x_n}$$

This shows that when the y -value of the outlier becomes extremely large, the estimated slope is mainly determined by the ratio y_n / x_n . The contribution of the other $n-1$ observations becomes negligible. This demonstrates how a single high-leverage observation can have a strong effect on the OLS regression estimate.